Introduction to Economics

Class 14 — Monopolistic competition; problems on Blocks 3–4

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Slide 1 Title

Block 4's last class. Class 12 had many sellers of an identical product and none of them able to move the price; class 13 had one seller that was the whole market. Most firms you buy from are neither: today's market has many sellers, each a little different from the others, and anyone free to open. The first half builds that case on one coworking space and ends with why such firms pay for adverts. The second half is practice: three graph problems on Blocks 3 and 4, a farm's costs and profit on a grid, two curves drawn from a table with a shift and an elasticity, and a price cap, each tried alone and then worked on the board the way you would work it on paper.

Quiz on Problem Set 3

15 minutes · pens down at 10:15

Use only your copy of Problem Set 3 and your own worked solutions.

Write your name and your copy number ("Copy NN", top right of your problem set).

Answers as numbers in the boxes; use the frames for your working.

Questions on both sides. Hand in the sheet; keep your problem set.

Quiz Start of class, on paper

A fifteen-minute quiz on Problem Set 3 opened the class, on one sheet printed on both sides, with your copy of the set and your own worked solutions at hand: you wrote your name and your copy number, then, for each problem, a result from your solutions and a short calculation with your copy's numbers after one change, answers in the boxes. Every answer was checked against your copy's variants. Problem Set 3 itself is not handed in: this quiz is its grade. Your result comes on Campus Virtual.

Plan for today

- Monopolistic competition: many sellers, each a little different
- A coworking space in the short run and the long run
- Why firms advertise
- Practice on the grid: a price taker's costs and profit
- Two curves from a table: a shift, the elasticity; a price cap

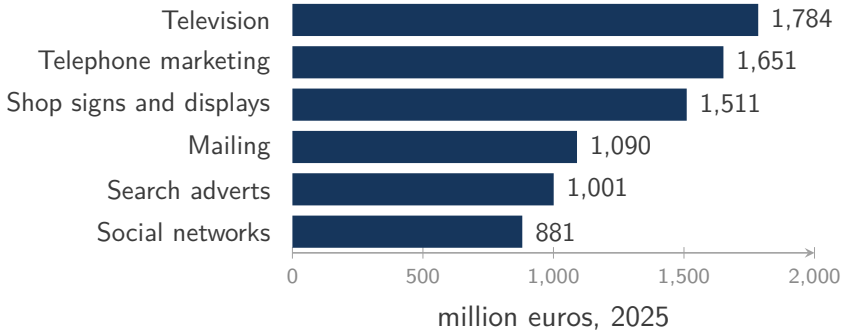
Slide 2 Plan for today

The class at a glance. Before the break, one idea built on one example: what is spent on adverts in Spain, which kind of market makes an advert worth paying for, a coworking space in the short run and after rivals open, and why it ends up with empty desks and a reason to advertise.

After the break, practice: three graph problems in the form of the practice exam on the course page, a price taker's costs and profit on a grid, two curves drawn from a table with a shift and an elasticity, and a price cap. You try each alone, then it is solved on the board the way you would solve it on paper. Problems like these are good practice for Midterm 1 on Thursday 19 November.

Spain's advertising: 12.7 billion euros in 2025

the six largest of 24 channels



About 260 euros for every person living in Spain.

InfoAdex, Estudio de la Inversión Publicitaria en España 2026 (2025 data): total 12,745.4 million euros, 0.76 % of gross domestic product (GDP); per person: ÷ about 49.6 million residents (Instituto Nacional de Estadística).

Slide 3 Spain's advertising: 12.7 billion euros in 2025

The hook: one real number about something nearly every firm does. In 2025, 12,745 million euros were spent on advertising in Spain, about three quarters of one per cent of everything the country produced. The figure is the estimate of InfoAdex, the company that measures advertising spending in Spain each year, in its 2026 study (2025 data); its 24 channels include telephone selling, signs and displays in shops and mail (these three alone, about a third of the total), not only adverts on screens. The 260 euros a person is that total divided by Spain's population of roughly 49.6 million (the INE's count), a division made for this class, not a figure of the study's.

Read it with care: it counts every advertiser, large brands and class 13's few big sellers included, and they advertise for more than one reason. The class derives one clear reason: a firm whose price is above its marginal cost gains from every extra customer, so winning one is worth paying for. The question carried to slide 8: why would a firm pay to win one more customer, when class 12's print shop, selling all it wants at the market price, would gain nothing from one?

Monopolistic competition: many sellers, each a little different, and anyone can open one

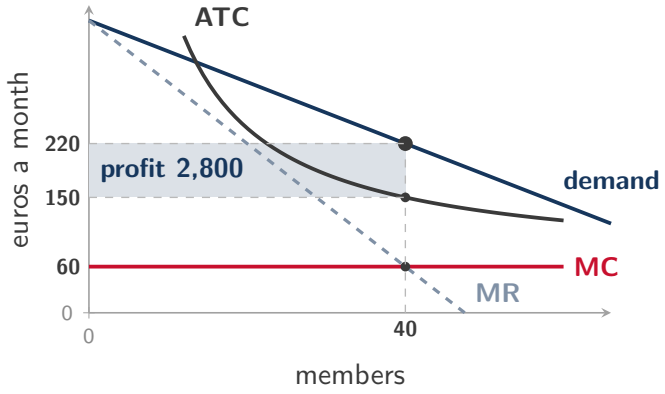
	Perfect competition (class 12)	Monopolistic competition (today)	Monopoly (class 13)
Sellers	many	many	one
Product	identical	each a little different	no close substitute
Entry	free	free	blocked
One firm's demand	flat at the price	slopes down	the whole market's

Price taker: sells all it wants at the market price (class 12, slide 6) · monopoly: the only seller, no close substitute (class 12, slide 20).

Slide 4 Monopolistic competition: many sellers, each a little different, and anyone can open one

The two outer columns are the last two classes; the middle column is today, and it borrows one feature from each. Monopolistic competition has many sellers, each selling a product that is at least a little different from the others (product differentiation: where it is, how it looks, how it is made, or simply how buyers see it), and free entry: anyone can open one or leave. The running example is coworking spaces in a city: each has its own location, light, coffee and people, and anyone can rent a floor and buy desks. Because its product differs, a space that raises its price loses some members but not all, so its own demand slopes down like a monopoly's: that is the "monopolistic" half. Because anyone can open, profits attract rivals, as under perfect competition: the "competition" half. The trap is to think that many sellers always means a price taker; that needs identical products as well.

In the short run a coworking space acts like a monopoly: 40 members at 220 euros, a profit of 2,800



60 desks
rent, furniture and the owner's time: 3,600 a month
each member costs 60 a month

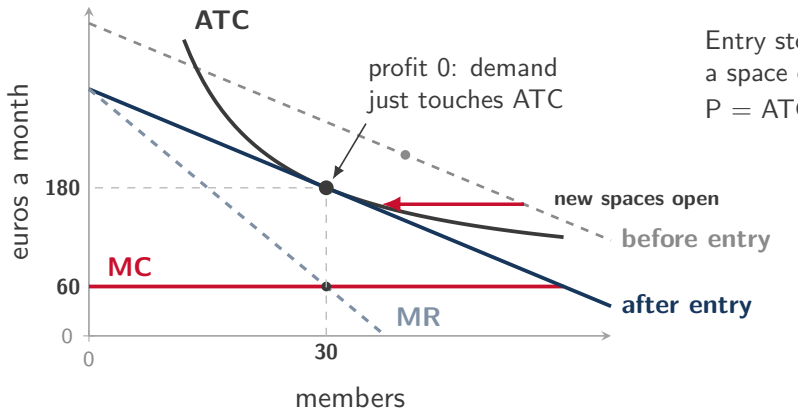
Like a monopoly:
sell while MR beats MC,
read the price off demand.

Marginal revenue (MR): what one more member adds to revenue (class 13, slide 4).
Average total cost (ATC): total cost ÷ members (class 11, slide 10) · MC: marginal cost.

Slide 5 In the short run a coworking space acts like a monopoly: 40 members at 220 euros, a profit of 2,800

One typical space, with the numbers in the card: 60 desks, 3,600 euros a month for rent, furniture and the owner's time whatever happens, and 60 more for each member (coffee, cleaning, electricity), its marginal cost. ATC is total cost per member, the 3,600 spread over the members plus 60; since each member costs the same 60, it falls until the desks are full: 180 at 30, 150 at 40, 120 with all 60 desks full. The demand slopes down, so marginal revenue lies below the price: to win one more member the space must lower the price for all. From 30 to 40 members revenue rises by 1,000 a month, 100 a member, more than the 60 each costs; from 40 to 50 by only 200, 20 a member. So, in the short run, it acts as a small monopoly: it stops at 40, reads the price off the demand curve (220, never off the MR line) and earns 70 a member above its average cost of 150, the shaded rectangle of 2,800 a month. The trap is to pick the most revenue: 50 members bring 9,000, but cost 600 more than 40 do for 200 more revenue.

New spaces open: each one's demand shifts left until profit is zero — 30 members at 180



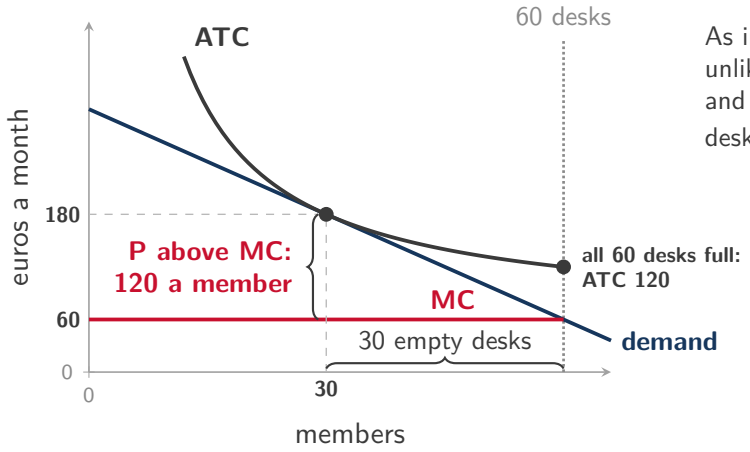
Entry stops when the best a space can do is zero profit: P = ATC.

Profit here is economic profit: every cost counted, the owner's own time included (class 11, slide 6).

Slide 6 New spaces open: each one's demand shifts left until profit is zero — 30 members at 180

A profit of 2,800 a month above every cost, the owner's time included, does not stay a secret: with free entry new spaces open. Each rival is a close substitute, so at every price our space keeps fewer members (at 220 a month, 20 instead of 40): its demand shifts left, which is not a move along the old curve. Its costs have not changed; the trap is to think entry lowers the space's costs, when it lowers the demand it faces while MC and ATC stay. The rule is the same: the new MR meets MC at 30 members and the new demand gives the price, 180; cost per member at 30 is 60 plus 3,600 divided by 30, also 180. Price equals ATC, so economic profit is zero: the owner's time and money earn exactly what they would earn elsewhere, nobody else wants to enter, nobody wants to leave. The new demand just touches ATC there and lies below it everywhere else, so at any other number of members the space would lose money (400 a month at 20 or 40). Losses would work the other way: spaces leave and the demand of the rest shifts right until profit is zero again.

Zero profit, yet the price is above MC and desks stand empty



As in class 12, P = ATC; unlike class 12, P > MC and ATC is still falling: desks stand empty.

Perfect competition, long run: price = lowest ATC (class 12, slide 18).

Fixed cost spread thinner, ATC falls (class 11, slide 15).

Slide 7 Zero profit, yet the price is above MC and desks stand empty

The long run in three facts: first, price equals ATC and profit is zero, because anyone can open, as in perfect competition. Second, the price is above marginal cost, 180 against 60, a markup of 120 a member, as in a monopoly, and it survives entry because the space's demand still slopes down: MR is below the price, so where the space stops, at MR = MC, the price is higher. Entry removed the profit, not the markup; the trap is to read zero profit as price equal to marginal cost. Third, excess capacity, 30 of the 60 desks empty: since each member costs the same 60, ATC falls until the desks are full, and filling them would bring the cost per member down to 120 but needs a lower price for every member; a downward-sloping demand can touch ATC only where ATC is still falling. It is partly waste, since members worth between 60 and 180 stay away, like the cable car's lost riders; but we also get variety, and no regulator could price every space: at 60 this one would not pay its rent. Class 13's line, more sellers push the price toward marginal cost, holds with a nuance: toward, not all the way.

With the price above MC, each extra member adds 120: so the space advertises

One more member: pays 180, costs 60 → adds 120 a month

A price taker: one more buyer at P = MC adds nothing

Critics

- shapes tastes, says little
- brand loyalty softens competition

Defenders

- tells prices, products, places
- helps new firms enter
- a costly campaign signals quality

A price taker produces while MC is below the price (class 12, slide 8) · 12.7 billion euros a year (slide 3).

Monopolistic competition: many sellers, each a little different, free entry.

Short run: like a monopoly. Long run: entry until profit is zero.

Even then the price stays above MC, and capacity is left unused.

Above MC, one more customer pays: that is why firms advertise.

Slide 8 With the price above MC, each extra member adds 120: so the space advertises

The hook's payoff: a member who joins the coworking space of slides 5–7 at 180 a month costs it 60, so each extra member adds 120 a month to profit even when profit as a whole is zero. Winning one is worth up to 120 a month, so the space pays for a sign or an advert; a price taker selling at marginal cost gains nothing from one more buyer, so sellers of identical products have little reason to advertise their own. This is one clear reason behind slide 3's 12.7 billion, not the whole of it: big firms and large brands spend part of that total, and firms advertise for several reasons; what they share is a price above marginal cost, without which an advert cannot pay.

Whether it is good for the rest of us, and whether brand names are, is a difficult question and a heated argument among economists. Critics: many adverts carry little information and work on our tastes, and the brand loyalty they build lets a brand charge more and makes entry harder. Defenders: adverts tell us prices, new products and where shops are, so markups shrink; a newcomer must be able to say it exists; and a firm that spends a fortune on a campaign is betting we will come back, so a costly advert can itself signal quality.

Slide 9 In one line

The first half in four lines, to read slowly: the first is slide 4, the second slides 5 and 6, the third slide 7, and the fourth, which answers slide 3's question, is slide 8.

Break

Back at 11:00

slide 10 of 16

Slide 10 Break

Ten minutes. After the break: practice, three graph problems, each tried alone and then solved on the board; have paper and a pen ready.

A pomegranate farm's costs: the table — four minutes alone

A farm sells crates of pomegranates at the market price of 30 euros a crate; it is a price taker. The table gives its total cost (TC) for 0 to 5 crates a day. [5.5 points; table cells as marked]

(a) Complete the table: the marginal cost (MC) of each crate (the extra cost of that crate) and the average total cost (ATC) at 2, 3 and 4 crates.

Crates a day	0	1	2	3	4	5
Total cost, TC (euros)	24	34	44	54	80	114
MC of that crate (euros) [0.5]	–	**10**	**10**	**10**	**26**	**34**
ATC (euros a crate) [1]	–	–	**22**	**18**	**20**	–

MC = change in TC: 34 − 24 = 10, 44 − 34 = 10, 54 − 44 = 10, 80 − 54 = 26, 114 − 80 = 34

ATC = TC ÷ crates: 44 ÷ 2 = 22, 54 ÷ 3 = 18, 80 ÷ 4 = 20

TC: total cost · MC: marginal cost, the extra cost of one more crate · ATC: average total cost, TC ÷ crates.

slide 11 of 16

Slide 11 A pomegranate farm's costs: the table — four minutes alone

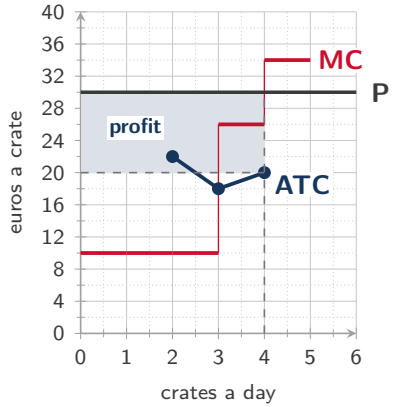
The first of two slides on one farm, worked as you would write it on paper. (1) MC, the extra cost of each crate, is the step in total cost from the column before: $34 - 24 = 10$ for the first crate, then $44 - 34 = 10$, $54 - 44 = 10$, $80 - 54 = 26$ and $114 - 80 = 34$, each written under its own crate. It stays at 10 for the first three crates (each takes the same extra work), then rises steeply. (2) ATC is total cost divided by the crates: $44 \div 2 = 22$, $54 \div 3 = 18$, $80 \div 4 = 20$.

(3) The 24 at zero crates is the fixed cost, paid even with no harvest: keep it in mind for the next slide. A right cell needs no working, but a light subtraction beside it costs nothing and catches a slip.

Slips: MC written as TC ÷ crates (that is ATC); every MC one column out of place (the second crate's 10 under the first); 34, the total cost itself, as the first crate's MC; ATC with the fixed cost left out ($20 \div 2 = 10$ at 2 crates is the average variable cost).

The farm on the grid: MC, ATC and profit — four minutes alone

The same farm, at the market price of 30 euros a crate. Its costs (euros):



[10.5 points; drawing 4.5; boxes as marked]

crates a day	0	1	2	3	4	5
TC	24	34	44	54	80	114
MC	–	10	10	10	26	34
ATC	–	–	22	18	20	–

(b) Draw MC as steps: over each crate, a horizontal line at that crate's MC. Mark the ATC at 2, 3 and 4 crates with dots and join them. Draw the price as a line labelled P.

(c) Best quantity (crates a day) [2] Profit (euros a day) [2]

4: the 5th costs 34 | **(30 − 20) × 4 = 40**

Shade the profit as a rectangle; write profit inside it.

(d) At 12 euros a crate: produce or shut down in the short run? Circle one; the reason in one line. [2]

(Produce) (Shut down) **12 is above the lowest AVC, 10 (1–3 crates): it loses 18, not the fixed 24**

MC: marginal cost · ATC: average total cost · AVC: average variable cost, variable cost ÷ crates · P: price.

Slide 12 The farm on the grid: MC, ATC and profit — four minutes alone

Worked as on paper. (1) Scale: one square is one crate across and 4 euros up, with dotted lines at the 2s: read the axis numbers first. (2) MC as steps: a flat line at 10 over the first three crates, at 26 over the fourth and at 34 over the fifth, labelled MC. ATC: dots at (2, 22), (3, 18) and (4, 20), joined and labelled. The price: a flat line at 30 across the grid, labelled P.

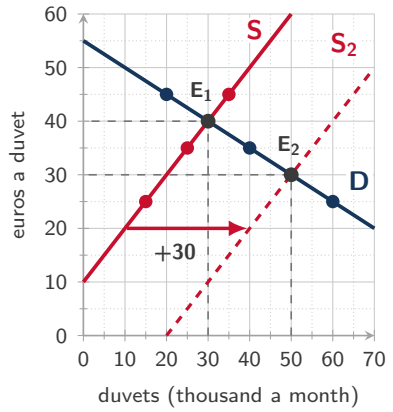
(3) Produce every crate whose MC is below the price: the fourth costs 26, the fifth 34, so 4 crates. Profit = (P − ATC) × crates = (30 − 20) × 4 = 40 euros a day: the rectangle from 0 to 4 crates between ATC (20) and P (30), with "profit" written inside.

(4) At 12 only the first three crates pay (MC 10): revenue 36, cost 54, a loss of 18. Shutting down still costs the fixed 24, so the farm produces: 12 is above the lowest average variable cost, (54 − 24) ÷ 3 = 10. The fixed cost is paid either way, so ATC never decides this.

Slips: the best quantity where MC or ATC is lowest; profit as P × Q; the rectangle down to the axis or up to MC; "12 < 18, shut down".

Duvets: two curves from a table, then a shift — six minutes alone

The market for duvets (quantities in thousand duvets a month). The table gives the quantities demanded and supplied at three prices; both curves are straight lines.



[12 points; drawing 6; boxes as marked]

Price (euros a duvet)	25	35	45
Quantity demanded (thousand a month)	60	40	20
Quantity supplied (thousand a month)	15	25	35

(a) Draw the demand curve and the supply curve through the table's points, label them D and S, and mark the equilibrium E_1.

(b) The price of polyester, an input used to make duvets, falls. At every price, sellers now offer 30 thousand more duvets a month than before. Draw the new supply curve, label it S_2, and mark the new equilibrium E_2.

	Price (euros) [1.5]	Quantity (thousand) [1.5]
(a) Before: E_1	**40 euros**	**30 thousand**
(b) After: E_2	**30 euros**	**50 thousand**

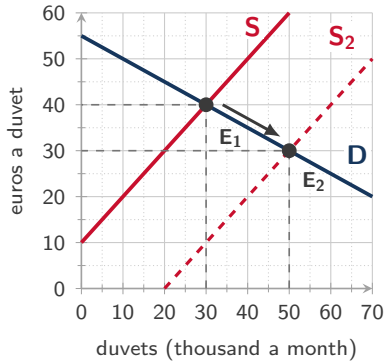
Slide 13 Duvets: two curves from a table, then a shift — six minutes alone

Worked as on paper. (1) Scale: one solid square is 10 (thousand duvets across, euros up); the dotted lines are the 5s, where the table's prices and the supplied quantities sit. (2) Demand: plot (60, 25), (40, 35) and (20, 45), the quantity across and the price up, and draw a straight line through them across the grid, labelled D. Supply: (15, 25), (25, 35), (35, 45), labelled S. They cross at E1: 40 euros and 30 thousand (check: at 40 buyers want 30 thousand and sellers offer 30).

(3) "30 thousand more at every price" moves every point of S 30 to the right: (15, 25) to (45, 25), (25, 35) to (55, 35). The line through them, S2, crosses D at E2: 30 euros and 50 thousand.

Supply shifted; demand did not. Buyers moved along D from E1 to E2: they buy more because duvets are cheaper, not because demand rose (slide 6's distinction, the other way round). The quantity rose by 20, not 30: at the lower price sellers offer less, a move down along S2. Slips: the price across; S2 drawn 30 up instead of 30 to the right (what a rise in cost would do); E2's quantity as 30 + 30 = 60.

The duvets: the elasticity from E_1 to E_2 — two minutes alone



[2 points; working shown]

E_1: 40 euros, 30 thousand · E_2: 30 euros, 50 thousand

(c) Demand did not move, so E_1 and E_2 are two points on the demand curve. Compute the price elasticity of demand between them by the midpoint method, and write it in the box as a positive number (a fraction or a decimal).

Elasticity (a positive number) [2]

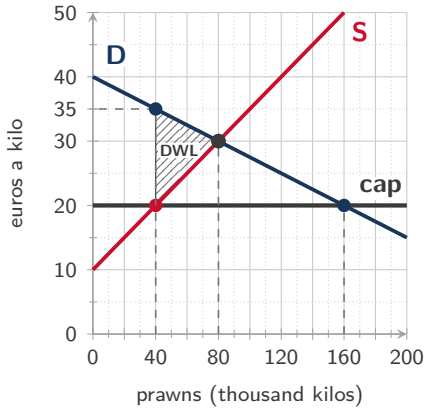
7/4 = 1.75

Working: **Quantity: 50 − 30 = 20; average (30 + 50) ÷ 2 = 40; 20 ÷ 40 = 1/2**
Price: 40 − 30 = 10; average (40 + 30) ÷ 2 = 35; 10 ÷ 35 = 2/7
(1/2) ÷ (2/7) = 1/2 × 7/2 = 7/4 = 1.75: demand is elastic here

E_1, E_2: the equilibrium before and after · price elasticity of demand: midpoint method, as a positive number.

Prawns at Christmas: a price cap — five minutes alone

The grid shows a city's market for prawns in Christmas week (quantities in thousand kilos): demand D and supply S. The city sets a maximum price (a price cap) of 20 euros a kilo.



[9 points; drawing 2; boxes as marked]

(a) Draw the cap as a line and label it cap.

(b) The quantity supplied, the quantity demanded and the shortage at the cap (thousand kilos).

Supplied [1.5] Demanded [1.5] Shortage [1]

Supplied	Demanded	Shortage
40	**160**	**160 − 40 = 120**

(c) Only the quantity supplied is sold, and it goes to the buyers who value it most. The deadweight loss (DWL) with the cap, in thousand euros; write DWL inside its area.

DWL [3]

300 **½ × (80 − 40) × (35 − 20) = 300 thousand euros**

Thousand kilos × euros a kilo = thousand euros. A triangle is ½ × base × height.

Slide 14 The duvets: the elasticity from E1 to E2 — two minutes alone

Demand did not move, so E1 and E2 are two points on one demand curve, and the elasticity between them says how strongly buyers react to the price. The midpoint method divides each change by the average of its two values. (1) Quantity: the change is $50 - 30 = 20$, the average $(30 + 50) \div 2 = 40$: $20 \div 40 = 1/2$. (2) Price: the change is $40 - 30 = 10$, the average $(40 + 30) \div 2 = 35$: $10 \div 35 = 2/7$. (3) Elasticity: the first result divided by the second, $(1/2) \div (2/7) = 1/2 \times 7/2 = 7/4 = 1.75$.

The price fell and the quantity rose, so the formula gives -1.75; it is reported as a positive number, 1.75. It is above 1, so demand is elastic between these points: from the midpoints, the quantity changes 1.75 times as much as the price, in percent. Write the three lines of working beside the box.

Slips: dividing by the starting values (from E1: $20 \div 30$ and $10 \div 40$, which gives 8/3; from E2: 6/5), when the averages give the same answer from either end; price over quantity (4/7, upside down); the slope of the line (1/2) instead of the elasticity; a minus sign in the box.

Slide 15 Prawns at Christmas: a price cap — five minutes alone

Worked as on paper. (1) Scale: across, one solid square is 20 thousand kilos and the dotted lines are the 10s; up, 10 euros a square, dotted 5s. Without the cap the market clears where S meets D, 80 thousand kilos at 30 euros. (2) The cap: a flat line at 20 across the grid, labelled cap; it is below 30, so it binds. (3) At 20 read S: sellers offer 40 thousand kilos; read D: buyers want 160. Shortage: $160 - 40 = 120$ thousand kilos, the queue at the fishmonger's.

(4) Only the 40 thousand offered are sold. The kilos from 40 to 80 thousand are lost, though buyers valued each above what it cost to bring in: the triangle between D and S from 40 to 80. Its left side, at 40, runs from S (20) up to D (35), a height of 15; its width is $80 - 40 = 40$: $\frac{1}{2} \times 40 \times 15 = 300$ thousand euros, with DWL written inside it.

Slips: the triangle's height taken up to 30, the old price (that gives 200), when its top corner is on D at 35; supplied and demanded swapped (sellers offer less at a lower price); the shortage measured from the old 80 ($160 - 80$ or $80 - 40$); the triangle drawn to the right of 80.

Before Monday 16 November

- Monday 16 November: review for Midterm 1 (Blocks 1–4), the full two hours
- Thursday 19 November: Midterm 1 (Blocks 1–4), in class
- Practise in Sextant: the units the course page lists



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Slide 16 Before Monday 16 November

Monday 16 November: the review session for Midterm 1, Blocks 1 to 4, the whole class. Thursday 19 November: Midterm 1, Blocks 1 to 4, in class. Problems like today's three are good practice: the practice exam on the course page has the same kind of pages, so do it in one sitting. Sextant has more, in the units the course page lists.

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